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$$\int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\cos x + \sin x} dx$$

$$\text{Stel } I = \int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\cos x + \sin x} dx$$

$$\text{Stel } x = \frac{\pi}{2} - t \Rightarrow dx = -dt$$

$$\text{Als } x = 0, \text{ dan } t = \frac{\pi}{2}$$

$$\text{Als } x = \frac{\pi}{2}, \text{ dan } t = 0$$

$$I = \int_{\frac{\pi}{2}}^0 \frac{\sin^3 \left(\frac{\pi}{2} - t \right)}{\cos \left(\frac{\pi}{2} - t \right) + \sin \left(\frac{\pi}{2} - t \right)} (-dt)$$

$$= \int_0^{\frac{\pi}{2}} \frac{\cos^3 t}{\sin t + \cos t} dt$$

$$= \int_0^{\frac{\pi}{2}} \frac{\cos^3 x}{\sin x + \cos x} dx$$

Dus:

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sin^3 x + \cos^3 x}{\sin x + \cos x} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{(\sin x + \cos x)(\sin^2 x - \sin x \cos x + \cos^2 x)}{\sin x + \cos x} dx$$

$$= \int_0^{\frac{\pi}{2}} (\sin^2 x - \sin x \cos x + \cos^2 x) dx$$

$$= \int_0^{\frac{\pi}{2}} (1 - \sin x \cos x) dx$$

$$= \int_0^{\frac{\pi}{2}} 1 dx - \int_0^{\frac{\pi}{2}} \sin x \cos x dx$$

$$= [x]_0^{\frac{\pi}{2}} - \left[\frac{\sin^2 x}{2} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2} - \frac{1}{2}$$

$$2I = \frac{\pi}{2} - \frac{1}{2}$$

$$I = \frac{\pi}{4} - \frac{1}{4} = \frac{\pi - 1}{4}$$

References